

Lokeshwaran Manohar

+492317554076 | lokeshwaran.manohar@tu-dortmund.de | [linkedin.com/in/Lokeshwaran](https://www.linkedin.com/in/Lokeshwaran) | github.com/Lokesh-26

Event-based robotic perception, learning-based 6D object pose estimation, and real-time vision for autonomous manipulation and logistics systems.

EDUCATION

Technical University of Dortmund

Master of Science in Automation and Robotics

October 2021 – October 2024

Dortmund, Germany

- **Master's thesis:** Object Localization and Tracking for High-Speed Robotic Grasping for Mobile Robots
- Relevant Courses: Robotic Manipulators, 3D Vision, Control Systems

Hindustan Institute of Technology and Science

Bachelor of Technology in Mechatronics

July 2017 – April 2021

Chennai, India

- **Bachelor's thesis:** Valkyrie - Design and Development of an Autonomous Bio-mimicking Robot and its Walking Gaits
- Relevant Courses: Robotics and Machine Vision, Control Systems, AI for Mechatronics System

EXPERIENCE

Research Assistant

Lehrstuhl für Förder- und Lagerwesen, TU Dortmund

February 2025 – Present

Dortmund, Germany

- Working on **event-based camera perception** for high-speed **object tracking** in robotic grasping and logistics scenarios.
- Implemented AI-driven dynamic pricing modules in the **SKALA** research project, supporting integration of scalable AI and blockchain technologies into cyber-physical production systems.
- Supporting system-level integration of mobile robots, perception modules, and real-time data processing in a multi-robot logistics environment.

Student Research Assistant

Lehrstuhl für Förder- und Lagerwesen, TU Dortmund (Advisor: M.Sc. Anas Gouda)

November 2022 – January 2025

Dortmund, Germany

- Developed and evaluated **6D pose estimation** pipelines using point clouds for robotic perception and grasping tasks.
- Supported dataset acquisition and preparation using the **Zivid** industrial 3D camera following the *BOP* format ([BOP](#)).
- Maintained mobile robots and assisted in digital twin and CPPS technology demonstrations.

Student Research Assistant

ANRO, Hindustan University (Advisor: Prof. Dr. Kuppan Chetty Ramanathan)

June 2019 – July 2019

Chennai, India

- Contributed to the development of a mobile wall-climbing robot for internal Non-Destructive Testing (NDT), improving safety and efficiency in industrial inspections.
- Designed 3D CAD models of the robot using Solidworks.

TECHNICAL COMPETENCIES

Programming Languages: Python, C++

Libraries & Frameworks: PyTorch, ROS, OpenCV, Open3D, CUDA, MATLAB, Simulink

Robotics & Systems: Mobile Robot Navigation, Robotic Manipulation, Perception Pipelines, Digital Twin Systems

Tools: Git, Docker, Linux, VS Code, PyCharm, SolidWorks, ANSYS, Webots

Research Methods: Experimental Design, Data Acquisition, Model Evaluation, Dataset Preparation

Scientific Writing: Research papers, Conference Publications, Technical Reports

Languages: English (Professional), German (B1), Tamil (Mother Tongue)

PROJECTS

Object Localization and Tracking for High-Speed Robotic Grasping for Mobile Robots [*Master's Thesis*]

Python, Pytorch, Linux, Git. Supervisor: Prof.'in Dr.-Ing. Alice Kirchheim

April 2024 – October 2024

- Developed an optimized data acquisition and preprocessing pipeline, reducing manual annotation effort and accelerating training for 6D pose estimation models.
- Benchmarked a custom dataset using the state-of-the-art **FoundationPose** 6D pose estimation model, evaluating accuracy and performance with the **BOP toolkit**.

- Designed a keypoint-based 6D pose estimation pipeline for known objects, improving inference speed for real-time robotic grasping applications.

Upper Body Control for the Soccer Playing Robot NAO

October 2022 – March 2023

C++, Matlab, Reinforcement Learning, Visual Studio, Git. Supervisor: Prof. Dr.-Ing. Uwe Schwiegelshohn

- Enhanced the autonomous walking capabilities of the **NAO humanoid robot** using **reinforcement learning**, improving locomotion performance for humanoid robotic applications.
- Implemented continuous walking actions using *Twin-Delayed Deep Deterministic policy gradient* (TD3).

Valkyrie - Design and Development of an Autonomous Bio-mimicking Robot and Its Walking Gaits [*Bachelor's Thesis*]

July 2020 – April 2021

SolidWorks, ANSYS, Matlab. Supervisor: Prof. Dr. Kuppan Chetty Ramanathan

- Designed and developed a bio-inspired **quadruped robot** capable of mimicking animal gait movements, contributing to advancements in robotic locomotion for uneven terrain and search-and-rescue missions.
- Achieved stability by calculating the support polygon, and structural analysis is performed in ANSYS software.

Autonomous Cost-Effective Robotic Exploration and Mapping System for Disaster Management

January 2020 – May 2020

Webots, Python, Matlab

- Developed autonomous behavior-based robots using heuristic methods for efficient deployment and navigation.
- Implemented advanced search algorithms for navigation, mapping, and path planning in disaster management scenarios.
- Simulated a swarm of robots in Webots to ensure system integrity and performance in various disaster situations.

PUBLICATIONS

- Anas Gouda, Shrutarv Awasthi, Christian Blesing, **Lokeshwaran Manohar**, Frank Hoffmann, and Alice Kirchheim. *MR6D: Benchmarking 6D Pose Estimation for Mobile Robots*. 2025. arXiv: 2508.13775 [cs.CV]. URL: <https://arxiv.org/abs/2508.13775>
- Taarun Srinivas, Adithya Krishna Karigiri Madhusudhan, **Manohar, Lokeshwaran**, Nikhit Mathew Stephen Pushpagiri, Kuppan Chetty Ramanathan, Mukund Janardhanan, and Izabela Nielsen. “Valkyrie—Design and development of gaits for quadruped robot using particle swarm optimization”. In: *Applied Sciences* 11.16 (2021), p. 7458. DOI: 10.3390/app11167458
- N Seenu, **Manohar, Lokeshwaran**, Nikhit Mathew Stephen, Kuppan Chetty Ramanathan, and MM Ramya. “Autonomous cost-effective robotic exploration and mapping for disaster reconnaissance”. In: *2022 10th International Conference on Emerging Trends in Engineering and Technology-Signal and Information Processing (ICETET-SIP-22)*. IEEE. 2022, pp. 1–6. DOI: 10.1109/ICETET-SIP-2254415.2022.9791800
- **Lokeshwaran, M.**, S.P. Nikhit Mathew, and A. Joshuva. “Raphael—The Smart Helmet”. In: *2020 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET)*. 2020, pp. 48–51. DOI: 10.1109/WiSPNET48689.2020.9198463

PATENT

An Autonomous Quadruped Robot System, October. 2022

Patent No: **202241058991**, *Intellectual Property of India*

ACADEMIC ACTIVITIES AND ACHIEVEMENTS

- Contributed to the organization of the BOP Challenge 2024 by assisting in HOPEv2 dataset data collection.
- Demonstrated robotic grasping using a 7-DoF KUKA manipulator at DORTBUNT and TU Dortmund Summer Festival 2024.
- Secured 12th position among 300 teams at Technoxian 2019 (RoboSoccer and Fastest Line Follower).